

Proper transposed sesqui arrays at all Sylvester–Hadamard powers

Carptopus
carptopus@163.com

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Abstract

For every power of two $t = 2^k$, $k \geq 2$, we construct a proper transposed sesqui array

$$SA^T(4t - 2, t, -, t - 1, t : (2t - 1) \times 2t).$$

The columns form the classical Sylvester–Hadamard trace design. We reduce the cell-ordering problem to a two-to-one finite-field map. Odd field dimensions are handled by an explicit inverse-pair map. In even dimension a Subiaco map and its zero fibre reduce the problem to two fixed algebraic curves. Geometric irreducibility, Artin–Schreier trace covers, and a Hasse–Weil estimate cover all even dimensions at least 34; fifteen exact polynomial-gcd certificates close the remaining even dimensions. A fixed resultant certificate proves properness, and an explicit array closes $t = 4$.

Status. The proof and its exact computations have passed implementation-separated internal checks. External mathematical review and an author-level priority inquiry remain pending, so all priority claims are qualified.

1 Introduction

A binary equireplicate row–column array is a transposed sesqui array when column–column and row–column concurrences are constant. It is proper when row–row concurrence is nonconstant. We study

$$(v, e, \lambda_{CC}, \lambda_{RC}; r, c) = (2q - 2, q/2, q/2 - 1, q/2; q - 1, q), \quad q = 2^d, \quad (1)$$

or, with $q = 2t$,

$$SA^T(4t - 2, t, -, t - 1, t : (2t - 1) \times 2t). \quad (2)$$

The columns below are classical affine trace hyperplanes. In the Sylvester case they agree block by block with the column incidence in the resolvable unordered triple-array construction of Gordeev and Öhman [5, Theorem 5.13]. That prior construction has constant row concurrence. The issue addressed here is the compatible ordered *proper* array.

The proof has five parts: a general two-fibre ordering theorem; an explicit odd-dimensional family; a Subiaco zero-fibre bridge in even dimension; a fixed-curve Hasse–Weil argument for all large even dimensions; and exact certificates for the finite complement. The computation is fixed and finite: two resultants, five irreducibility and trace checks, fifteen finite-field witnesses, and one boundary array. It contains no array classification or extrapolation from sampled dimensions.

To the best of our knowledge, (2) is the first proper construction for every power of two t . The $t = 8$ instance fills the proper entry marked $?, T$ by Jäger et al. [7, Table 2]; T records a triple array rather than a proper one.

2 Trace columns and a two-fibre ordering theorem

Let $F = \text{GF}(q)$, $q = 2^d$, and let $\text{Tr} : F \rightarrow \mathbb{F}_2$ be the absolute trace. Index symbols by $(h, b) \in F^* \times \mathbb{F}_2$. For $c \in F$, set

$$\mathcal{B}_c = \{(h, \text{Tr}(ch)) : h \in F^*\}. \quad (3)$$

Lemma 2.1. *For $c \neq c'$, $|\mathcal{B}_c \cap \mathcal{B}_{c'}| = q/2 - 1$.*

Proof. A common symbol satisfies $\text{Tr}((c + c')h) = 0$. The nondegenerate trace pairing gives a hyperplane of size $q/2$; remove $h = 0$. $\square$

Theorem 2.2 (Two-fibre ordering). *Suppose $f : F \rightarrow F^*$ is exactly two-to-one with image Y , the map $x \mapsto xf(x)$ permutes F^* , and every fibre $\{x, x'\}$ satisfies*

$$\text{Tr}((x + x')f(x)) = 1. \quad (4)$$

For rows $r \in F^$, columns $c \in F$, define*

$$A(r, c) = (rf(cr), \text{Tr}(crf(cr))). \quad (5)$$

Then A is a binary equireplicate transposed sesqui array with (1), and for distinct rows

$$|A_r \cap A_s| = 2|Y \cap (s/r)Y|. \quad (6)$$

Proof. For $c = 0$, the first coordinate $rf(0)$ runs through F^* . For $c \neq 0$, putting $x = cr$ gives first coordinate $xf(x)/c$, which also runs through F^* . Thus column c is $\mathcal{B}_c$.

In a fixed row, the two inputs in a fibre of value y give first coordinate ry . Condition (4) gives opposite binary coordinates, so the row contains both symbols over every member of rY , with no repetition. A fixed first coordinate h is supported by the $q/2$ rows $r = h/y$, $y \in Y$; hence each symbol has replication $q/2$. A row and column share one symbol over each member of rY , so their intersection is $q/2$. Two rows share both symbols exactly over $rY \cap sY$, proving (6). $\square$

3 Odd field dimension

Assume d is odd and define

$$f(0) = f(1) = 1, \quad f(x) = \frac{1}{x + x^{-1}} = \frac{x}{x^2 + 1} \quad (x \notin \{0, 1\}). \quad (7)$$

If $H = \ker \text{Tr}$, then

$$Y = (H \setminus \{0\}) \cup \{1\}. \quad (8)$$

Indeed, $f(x) = y$ reduces through $u = xy$ to $u^2 + u = y^2$. It has two roots precisely for nonzero $y \in H$, and the roots in x are mutual inverses. The special fibre $\{0, 1\}$ has image one and is disjoint from the ordinary images because $\text{Tr}(1) = 1$. Every fibre satisfies (4). Moreover,

$$xf(x) = \left(\frac{x}{x+1} \right)^2$$

on $F^* \setminus \{1\}$, so the required permutation follows from a Möbius map and Frobenius.

For $a \in F^* \setminus \{1\}$,

$$|Y \cap aY| = q/4 - 1 + \mathbf{1}_{\text{Tr}(a)=0} + \mathbf{1}_{\text{Tr}(a^{-1})=0}. \quad (9)$$

Writing

$$K_q = \sum_{a \in F^*} (-1)^{\text{Tr}(a+a^{-1})},$$

elementary character sums give

$$N_{00} = (q - 3 + K_q)/4, \quad N_{11} = (q + 1 + K_q)/4.$$

The binary Kloosterman bound $|K_q| \leq 2\sqrt{q}$ [3] gives $N_{00} > 0$ and $N_{11} > 1$ for $q \geq 32$. Thus (9) assumes at least two values.

Theorem 3.1. *For every odd $d \geq 5$, (5) and (7) give a proper array with (1).*

4 The Subiaco bridge in even dimension

Let $d \geq 4$ be even. Call $c \in F$ admissible when

$$c \neq 0, \quad c^2 + c + 1 \neq 0, \quad \text{Tr}(c^{-1}) = 1. \quad (10)$$

Put

$$\begin{aligned} u &= 1 + c + c^2, & D(x) &= x^2 + cx + 1, \\ N_F(x) &= c^2(x^4 + x) + c^2u(x^3 + x^2), \end{aligned}$$

and take the normalized Subiaco o-polynomial

$$F_c(x) = \frac{N_F(x)}{D(x)^2} + x^{1/2}, \quad B_c(x) = F_c(x) + ux. \quad (11)$$

The denominator has no field root under (10): if $D(x) = 0$, then $y = x/c$ would satisfy $y^2 + y = c^{-2}$, whereas $\text{Tr}(c^{-2}) = \text{Tr}(c^{-1}) = 1$. We make the finite-geometric input and the additional algebra explicit. Put

$$\eta = c^2 + c^5 + c^{1/2}, \quad \kappa = \frac{\eta}{cu},$$

Here $u \neq 0$, and also $\eta \neq 0$. For the latter claim, suppose $\eta = 0$ and put $w = c^{-1/2}$. Then

$$w^9 + w^6 + 1 = (w^3 + w + 1)(w^6 + w^4 + w^2 + w + 1) = 0.$$

The two displayed factors are irreducible over $\mathbb{F}_2$, of degrees three and six. In the cubic case, $3 \mid d$; since d is even, the extension multiplier $d/3$ is even, so $\text{Tr}_F(w^2) = 0$. In the sextic case, $6 \mid d$, and the zero w^5 -coefficient gives $\text{Tr}_{\mathbb{F}_{64}/\mathbb{F}_2}(w) = 0$, hence again $\text{Tr}_F(w^2) = 0$. Both cases contradict $\text{Tr}(c^{-1}) = \text{Tr}(w^2) = 1$.

The exact coordinate formulas below are recorded by O’Keefe–Thas [12, pp. 178–179], with their d equal to our c and their scalar a equal to our κ :

$$N_G(x) = c^4x^4 + c^3(1 + c^2 + c^4)x^3 + c^3(1 + c^2)x,$$

and

$$G_c(x) = \frac{N_G(x)}{\eta D(x)^2} + \frac{c^{1/2}x^{1/2}}{\eta}. \quad (12)$$

The Subiaco construction [4] supplies that F_c is an o-polynomial and that, in the normalized q -clan convention [2, Eq. (19)], the matrices

$$A_x = \begin{pmatrix} F_c(x) & x^{1/2} \\ 0 & \kappa G_c(x) \end{pmatrix} \quad (x \in F)$$

form a q -clan. Thus for $s \neq t$ the binary quadratic form attached to $A_s + A_t$ is anisotropic. Its cross coefficient has square $s + t$, so the characteristic-two anisotropy criterion gives exactly

$$\text{Tr} \left(\kappa \frac{(F_c(s) + F_c(t))(G_c(s) + G_c(t))}{s + t} \right) = 1. \quad (13)$$

Here $F_c(0) = 0$ and $F_c(1) = 1$, so the first fact and the o-polynomial/two-to-one criterion [9, Theorem 2.2] imply that $B_c(x) = F_c(x) + ux$ is exactly two-to-one.

Lemma 4.1 (Subiaco fibres). *Every fibre $\{s, t\}$ of B_c satisfies*

$$\text{Tr}((s + t)/c) = 1. \quad (14)$$

Proof. Define

$$P(x) = c^2 x(x + c), \quad H(x) = \frac{P(x)}{D(x)} + (u + c^{-1/2})x^{1/2}.$$

Direct expansion in characteristic two gives

$$N_G(x)/c + uN_F(x) = P(x)^2 + P(x)D(x),$$

and hence

$$\kappa u G_c(x) + x/c + u F_c(x) + u^2 x = H(x)^2 + H(x). \quad (15)$$

Taking absolute traces in (15) yields

$$\text{Tr}(\kappa u G_c(x) + x/c) = \text{Tr}(u B_c(x)).$$

If $B_c(s) = B_c(t)$, addition at s, t gives

$$\text{Tr}(\kappa u (G_c(s) + G_c(t)) + (s + t)/c) = 0.$$

But $F_c(s) + F_c(t) = u(s + t)$, so (13) gives $\text{Tr}(\kappa u (G_c(s) + G_c(t))) = 1$. The last two identities prove (14). $\square$

Lemma 4.2 (Exterior-line bridge). *Let $I = B_c(F)$. For a fibre $\{a, a'\}$, put $b = B_c(a)$, $\delta = a + a'$. If*

$$u\delta \notin I + b, \quad (16)$$

then Theorem 2.2 applies to

$$x_z = \frac{z + a}{B_c(z) + b}, \quad f(x_z) = c^{-1}(B_c(z) + b) \quad (17)$$

for $z \notin \{a, a'\}$, completed by $f(0) = f(u^{-1}) = c^{-1}u\delta$.

Proof. Set

$$r_a(z) = \frac{F_c(z) + F_c(a)}{z + a}, \quad r_a(a) = 0.$$

The secant-slope property of an o-polynomial says that r_a permutes F , and

$$B_c(z) + b = (z + a)(r_a(z) + u).$$

The partner a' is the unique point with $r_a(a') = u$. Hence the x_z in (17) run through $F \setminus \{0, u^{-1}\}$. Every nonzero member of $I + b$ gives one ordinary two-element label fibre, while (16) prevents the special label from colliding with an ordinary one. Thus f is exactly two-to-one.

For ordinary inputs,

$$x_z f(x_z) = c^{-1}(z + a).$$

These products cover $F \setminus \{0, c^{-1}\delta\}$; the two special inputs give the missing products $0, c^{-1}\delta$. Thus $xf(x)$ permutes F . Finally, an ordinary fibre coming from $B_c(s) = B_c(t)$ satisfies (4) by Lemma 4.1, while the special fibre satisfies it because $\text{Tr}(\delta/c) = 1$ for the anchor pair a, a' . $\square$

5 The zero fibre and a fixed properness certificate

Write $c = h^2$. The fibre of B_c containing zero is

$$\{0, cr^2\}, \quad \text{Tr}(r) = 1, \quad (18)$$

where the nonzero partner satisfies

$$\begin{aligned} K_h(R) &= 1 + (h + h^3)R + (h^7 + h^9 + h^{11})R^3 + h^8 R^4 \\ &\quad + h^{11} R^7 + h^8 R^8 + (h^9 + h^{11} + h^{13})R^9 = 0. \end{aligned} \quad (19)$$

The exterior condition becomes

$$ucr^2 \notin I. \quad (20)$$

For $Y \subset F^*$, put $S_j(Y) = \sum_{y \in Y} y^j$. If $|Y| = q/2$ and every nontrivial $|Y \cap aY|$ equals $q/4$, then Y is a cyclic $(v, k, \lambda) = (q - 1, q/2, q/4)$ difference set. Apply the multiplier theorem to its complement $D = F^* \setminus Y$, whose parameters are

$$(v, k', \lambda', n) = (q - 1, q/2 - 1, q/4 - 1, q/4).$$

Here n is a power of the prime 2, $n > \lambda'$, and $\gcd(2, v) = \gcd(2, q - 1) = 1$. The prime-power case of the Second Multiplier Theorem [14, Section 4] therefore makes 2 a numerical multiplier of D , and hence also of Y . Thus $Y^2 = \alpha Y$ for some $\alpha \in F^*$. When $S_1(Y) \neq 0$, summing and normalizing gives

$$S_j(Y) \in \{0, S_1(Y)^j\} \quad \text{for all } j. \quad (21)$$

Direct trace-hyperplane summation gives, for $I = B_c(F)$,

$$A := \sum_{v \in I} v = cu, \quad (22)$$

$$\begin{aligned} C := \sum_{v \in I} v^3 &= h^3 + h^7 + h^9 + h^{10} + h^{11} + h^{12} + h^{16} \\ &= h^3(h^2 + h + 1)(h^3 + h + 1)^2(h^5 + h^4 + h^2 + h + 1). \end{aligned} \quad (23)$$

Here every factor is nonzero under (10). Indeed $h \neq 0$, $(h^2 + h + 1)^2 = c^2 + c + 1 = u \neq 0$, and the remaining cubic and quintic factors are irreducible over $\mathbb{F}_2$. If either vanished in $F = \mathbb{F}_{2^d}$, then its degree $m \in \{3, 5\}$ would divide d . Since d is even, d/m would be even, and every element of $\mathbb{F}_{2^m}$ would have absolute trace zero in F ; applied to $h^{-2} = c^{-1}$, this contradicts (10). For the zero fibre the unscaled label set is

$$X = (I \setminus \{0\}) \cup \{ucr^2\}.$$

Since d is even and $\text{Tr}(r) = 1$, one has $r \neq 1$; hence $S_1(X) = cu(1 + r^2) \neq 0$, as required when applying (21) below.

Lemma 5.1 (Computer-assisted fixed resultant). *For every even $d \geq 4$ and admissible $c = h^2$, the zero fibre cannot satisfy $S_3(X) = 0$ or $S_3(X) = S_1(X)^3$.*

Proof. Eliminating R between (19) and the two degeneracy equations gives

$$h^{54}(h+1)^2(h^2+h+1)^{14}R_{21}(h)R_{109}(h) \quad (24)$$

and

$$h^{54}(h+1)^2(h^2+h+1)^{14}R_{26}(h)R_{29}(h)R_{75}(h). \quad (25)$$

The five R_m are monic irreducible polynomials over $\mathbb{F}_2$. Their degrees and inverse-square root traces are

factor	R_{21}	R_{109}	R_{26}	R_{29}	R_{75}
degree	21	109	26	29	75
$\text{Tr}(\alpha^{-2})$	0	0	0	1	0

The low-degree factors contradict admissibility. Each trace-zero factor does so in every ambient extension. If $R_{29}(h) = 0$ in F_{2^d} , then $29 \mid d$; because d is even, the extension degree $d/29$ is even, so its minimal trace one becomes ambient trace zero. $\square$

The supplement gives the complete exponent lists. One verifier uses symbolic Bareiss resultants, Rabin tests, and quotient-field traces. A code-independent verifier reconstructs both resultants from all 256 evaluations in an AES model of $\text{GF}(256)$, using Gaussian Sylvester determinants, interpolation, Berlekamp nullities, and coefficient traces. Independent degree bounds 240 and 214 are below 256; both reconstructed resultants have degree 214.

Corollary 5.2. *An admissible zero fibre satisfying (20) gives a proper array with (1).*

Proof. The exterior bridge gives Theorem 2.2. Equation (22), (23), Lemma 5.1, and (21) force nonconstant multiplicative intersections. $\square$

6 Fixed curves for the exterior condition

Put

$$a = c = h^2, \quad t = hr, \quad U(a) = 1 + a + a^2,$$

and define

$$K(a, T) = 1 + (1 + a)T + a^2U(a)T^3 + a^2T^4 + a^2T^7 + T^8 + U(a)T^9, \quad (26)$$

$$D(a, P) = P^8 + a^2P^4 + 1, \quad (27)$$

$$L(a, T, P) = PK(a, P) + U(a)T^2D(a, P). \quad (28)$$

For admissible a , the denominator in (29) below never vanishes on F . In fact

$$D(a, p) = (p^4 + ap^2 + 1)^2.$$

If the factor on the right vanished, then $a \neq 0$ and $y = p^2/a$ would satisfy $y^2 + y = a^{-2}$, impossible because $\text{Tr}(a^{-2}) = \text{Tr}(a^{-1}) = 1$. Direct substitution gives

$$B_a(aw^2) = \frac{pK(a, p)}{D(a, p)}, \quad p = hw. \quad (29)$$

For admissible a , $K(a, T)$ has a unique field root t , the nonzero partner of the zero fibre.

Lemma 6.1. *The zero fibre is exterior if and only if $L(a, t, P)$ has no root in F .*

Proof. By (29), the forbidden membership $U(a)t^2 \in B_a(F)$ is equivalent to a field root p of $L(a, t, P)$. $\square$

Let C normalize the projective closure of $K(a, t) = 0$, and let X normalize that of $K(a, t) = L(a, t, p) = 0$.

Proposition 6.2. *The curves C and X are geometrically integral over $\mathbb{F}_2$, and so are the Artin–Schreier covers*

$$y^2 + y = 1/a \quad (30)$$

on both curves.

Proof. The Newton polygon of K , as a polynomial in a over $\mathbb{F}_2((t))$, has coefficient valuations $0, 1, 3, 3, 3$ and one lower edge from $(0, 0)$ to $(4, 3)$. In a factorization, lower Newton polygons add. Every nonzero lower edge would therefore have slope $3/4$, hence horizontal length divisible by 4; the total horizontal length is 4. Thus no two positive-degree factors exist, over any algebraic extension of the constant field, and C is geometrically integral.

Put $t = 1 + s$. The lowest homogeneous part at $(a, s) = (0, 0)$ is $a^2(s + a)$. The tangent $s + a = 0$ occurs with multiplicity one and gives a branch on the normalization with $v(a) = 1$. After base change to $\overline{\mathbb{F}}_2$, the function $1/a$ has an odd pole and cannot be of the form $g^2 + g$. Hence the Artin–Schreier cover of C is geometrically integral.

At the smooth rational point $Q = (1, 1)$, one has $U(1) = 1$, and the monic polynomial L/U specializes to

$$P^{10} + P^9 + P^5 + P + 1, \quad (31)$$

which is irreducible and separable over $\mathbb{F}_2$. If L/U factored over $\mathbb{F}_2(C)$, choose monic factors. Their coefficients are integral over the DVR $\mathcal{O}_{C, Q}$ and lie in its fraction field, hence belong to the DVR because it is integrally closed. Reduction at Q would give a nontrivial monic factorization of (31). Hence the fibre product is integral and degree ten over C .

The irreducible specialization gives a smooth closed point of degree ten. A second smooth closed point has degree three:

$$(a, t, p) = (\alpha, \alpha + 1, 1), \quad \alpha^3 + \alpha + 1 = 0,$$

as the two gradients are independent. For the proper normalization X , the degree of its full constant field divides every closed-point degree. It therefore divides both 10 and 3, hence equals one. Since $\mathbb{F}_2$ is perfect and the function-field extension is separable, X is smooth with exact constant field $\mathbb{F}_2$, and therefore geometrically integral.

Finally,

$$L(0, 1, P) = (P + 1)^8(P^2 + P + 1).$$

At a root $\omega^2 + \omega + 1 = 0$, one has $\partial L / \partial P = \omega \neq 0$. It lifts unramified above the branch with $v(a) = 1$, so $1/a$ again has an odd pole. This proves the assertion for the cover of X . $\square$

7 Hasse–Weil existence for large even dimension

We first make the ramification estimate explicit. For $Z = C$ or X , let $n_Q > 0$ be the order of a zero of a at a geometric point Q . In the local Artin–Schreier class, successive replacements by $g^2 + g$ either remove the pole or reduce it to an odd order $m_Q \leq n_Q$. Such a pole has different exponent $m_Q + 1 \leq 2n_Q$ [15, Corollary 2.3]. Thus the total different has degree at most $2 \deg(a : Z \rightarrow \mathbf{P}^1)$, and Riemann–Hurwitz [16, Theorem 3.4.13] gives

$$g(Y_Z) \leq 2g(Z) - 1 + \deg(a : Z \rightarrow \mathbf{P}^1).$$

The plane degree bound for C and the preceding inequality give

$$g(C) \leq 45, \quad g(Y_C) \leq 98, \quad (32)$$

because K has total degree at most 11 and $\deg(a : C \rightarrow \mathbf{P}^1) \leq 9$. The fibre product has projective degree at most $11 \cdot 12 = 132$. Castelnuovo’s genus bound [6, Ch. IV, Theorem 6.4], used here only in the deliberately weaker plane-form estimate $g \leq (D - 1)(D - 2)/2$, gives

$$g(X) \leq 8515, \quad g(Y_X) \leq 17119, \quad (33)$$

using $\deg(a : X \rightarrow \mathbf{P}^1) \leq 90$.

Remove $a = 0$, poles of a , and $U(a) = 0$, and denote the resulting open curves by C° and X° . The deleted divisors have total degree at most 36 on C , 360 on X , 72 on Y_C , and 720 on Y_X . For either open curve Z° , let $N_0(Z^\circ)$ count points with $\text{Tr}(1/a) = 0$. The equation $y^2 + y = 1/a$ has exactly two rational solutions over such a point and none when the trace is one. Consequently

$$\#Y_Z^\circ(\mathbb{F}_q) = 2N_0(Z^\circ), \quad N_1(Z^\circ) = \#Z^\circ(\mathbb{F}_q) - \frac{1}{2}\#Y_Z^\circ(\mathbb{F}_q).$$

Let $\tilde{A}_q = N_1(C^\circ)$ and $\tilde{T}_q = N_1(X^\circ)$ be the corresponding counts on the normalizations. Applying Hasse–Weil [16, Theorem 5.2.3] to the four smooth projective curves gives

$$\tilde{A}_q \geq (q + 1)/2 - 188\sqrt{q} - 36, \quad (34)$$

$$\tilde{T}_q \leq (q + 1)/2 + 34149\sqrt{q} + 360. \quad (35)$$

We now account explicitly for the fact that these are normalization-point counts. For any set S of rational points, put $m_P = \#\nu^{-1}(P)(\mathbb{F}_q)$, where ν is the normalization. Then $\#\nu^{-1}(S)(\mathbb{F}_q) = \sum_{P \in S} m_P$. At a smooth point $m_P = 1$; at a singular point $0 \leq m_P \leq r_P$, where r_P is the number of geometric branches. Hence

$$\#\nu^{-1}(S)(\mathbb{F}_q) - \#S \leq \sum_{P \in \text{Sing}} (r_P - 1).$$

For the degree-at-most-11 geometrically integral plane curve, the normalization exact sequence

$$0 \longrightarrow \mathcal{O}_{\bar{C}} \longrightarrow \nu_* \mathcal{O}_C \longrightarrow \mathcal{Q} \longrightarrow 0$$

gives $\sum_P \delta_P = \text{length } \mathcal{Q} = p_a(\bar{C}) - g(C) \leq 45$. At a point with r_P geometric branches, reduction of the semilocal normalization modulo its maximal ideals induces a surjection $(\nu_* \mathcal{O}_C / \mathcal{O}_{\bar{C}})_P \rightarrow \mathbb{F}_2^{r_P} / \mathbb{F}_2$; hence $r_P - 1 \leq \delta_P$. Applying the displayed inequality to the admissible trace-one affine pairs gives

$$A_q \geq \tilde{A}_q - 45.$$

The same fibre identity in the reverse direction gives $\#S - \#\nu^{-1}(S)(\mathbb{F}_q) \leq \#(S \cap \text{Sing})$, because only a singular rational point can have no rational point above it. For an admissible rational triple, direct differentiation gives $\partial L/\partial p = D(a, p) \neq 0$. If its base pair is smooth on $K = 0$, then $\nabla K = (K_a, K_t, 0)$ and $\nabla L = (L_a, L_t, D)$ are independent, so the triple is smooth and lifts uniquely to a rational normalization point. Hence a rational bad triple with no rational lift can occur only above a singular base pair. There are at most 45 such pairs, and the admissible polynomial $L(a, t, p)$ has degree ten in p , so allowing all of them separately gives

$$T_q \leq \tilde{T}_q + 10 \cdot 45 = \tilde{T}_q + 450,$$

where T_q is now the number of distinct admissible bad affine triples.

By the exact two-to-one parameterization, the nonvanishing of $D(a, p)$, and the bijection $p = hw$, every bad parameter contributes exactly two p -values and every good one contributes none. Therefore

$$G_q \geq A_q - T_q/2 \geq (q+1)/4 - 17262.5\sqrt{q} - 486. \quad (36)$$

This is positive at $q = 2^{34}$ and remains positive thereafter.

Theorem 7.1. *For every even $d \geq 34$, a proper array with (1) exists.*

8 The finite even basis

For each remaining even d , represent $F_{2^d} = \mathbb{F}_2[X]/(m_d)$. Integers encode binary polynomial coefficients, including the leading bit.

d	modulus	a	t
4	19	2	9
6	97	6	58
8	501	15	250
10	1627	3	65
12	7993	26	3012
14	30279	3	6853
16	124427	3	49789
18	320757	26	159424
20	1706235	9	1018552
22	5740747	6	1921895
24	33394861	6	9911191
26	99611673	3	53205065
28	336533213	3	177941084
30	1179354645	3	252669766
32	7185006415	8	164093156

For each row, exact polynomial arithmetic verifies that m_d is irreducible,

$$aU(a) \neq 0, \quad \text{Tr}(a^{-1}) = 1,$$

$$\deg \gcd(K(a, T), T^{2^d} - T) = 1 \quad (37)$$

with unique root the displayed t , and

$$\gcd(L(a, t, P), P^{2^d} - P) = 1. \quad (38)$$

Repeated Frobenius reduction proves these identities without enumerating the field. An independent implementation obtains the same degrees.

Theorem 8.1. *For every even d , $4 \leq d \leq 32$, a proper array with (1) exists.*

9 The all-power theorem

For $d = 3$, $t = 4$, the following 7×8 array on symbols $0, \dots, 13$ is a certificate:

0	1	3	12	2	10	11	13
1	11	0	2	4	13	12	9
11	0	2	8	1	5	13	12
3	10	9	0	13	6	4	7
6	7	12	10	3	1	5	8
9	5	7	4	8	2	6	11
8	4	5	6	7	9	3	10

Every symbol occurs four times, with $CC = 3$ and $RC = 4$. The row-concurrence multiset is

$$\{2^{(6)}, 4^{(9)}, 6^{(6)}\}.$$

The fourteen symbol-incidence vectors occur in seven complementary pairs. Choosing from each pair the vector absent from the first column and adjoining the zero vector gives a three-dimensional binary linear space whose seven nonzero vectors all have weight four. Thus its column support is the Sylvester trace design at $q = 8$.

Theorem 9.1 (All Sylvester–Hadamard powers). *For every $k \geq 2$, $t = 2^k$, there exists*

$$SA^\top(4t - 2, t, -, t - 1, t : (2t - 1) \times 2t).$$

Proof. Set $d = k + 1$, $q = 2^d = 2t$. The displayed array covers $k = 2$. For even $k \geq 4$, d is odd and Theorem 3.1 applies. For odd $k \geq 3$, d is even and Theorem 7.1 or Theorem 8.1 applies. $\square$

10 Prior-work boundary

The trace columns (3) are the classical Sylvester–Hadamard design. With

$$A_h = \{c \in F^* : \text{Tr}(ch) = 0\},$$

the paired supports $A_h \cup \{0\}$ and $F^* \setminus A_h$ are exactly those in Gordeev–Öhman [5, Theorem 5.13]. Their result gives an exact-parameter resolvable unordered triple array but not the proper row supports or ordering in Theorem 2.2.

The deletion–exchange construction of Jäger et al. [8, Construction 5.2 and Theorem 5.3] is an especially close prior framework. Applied to a Youden rectangle with $(n, k, \lambda) = (4t - 1, 2t - 1, t - 1)$, it gives exactly the dimensions, replication, and column concurrence in (2). What it guarantees automatically is the binary equireplicate mono-array layer; constant row–column concurrence and properness require an additional compatibility. The contribution here is not the deletion–exchange transformation, but a compatible proper ordering for every power-of-two t , supplied by Theorem 2.2 and the maps constructed in the preceding sections.

Same-parameter triple arrays were also known. McSorley et al. [10] give the $t = 8$ parameters, and Preece et al. [13] give the same form when $2t - 1$ is an odd prime power. Their row concurrence is constant. O-polynomials, Subiaco hyperovals, line ovals, and the two-to-one relation for $O(x) + ux$ are also prior theory [4, 9, 1].

For the proper object, Jäger et al. [7, Table 2] record $(v, r, c) = (30, 15, 16)$ as $?, T$: their search and listed constructions did not produce a proper transposed sesqui array, a triple array was known, and nonexistence was not proved. The $t = 8$ instance here addresses that recorded proper-existence gap; this status statement alone does not establish global priority.

The recent near-triple-array framework of Gordeev–Markström–Öhman [11] includes new constructions and selected enumerations of proper row–column designs, but its listed proper-design cases do not include the 15×16 , 30-symbol target or the all-power family here.

Thus the qualified novelty claim is only the proper ordered all-power family: it is not the first Hadamard support, same-parameter unordered design, or same-parameter triple array.

11 Reproducibility

The supplement provides both resultant verifiers, all symbolic identities, the geometry checks, fifteen finite-basis certificates, the boundary array, a single full-chain command, and a SHA-256 manifest. The independent resultant coefficient hashes are

```
S3=0: 258eeeca81b726d3bc13d4ab37edfdc0c118ec5b0e813c7dec46ef9ecf7f241
S3=S1^3: 212ed4eed451dbc3f7d078b3a61ffdbf51bceb117f548f3febfa68866d032e59
```

The manuscript is internally proof-frozen but externally unreviewed.

Declaration of AI-assisted research and writing

This research program used OpenAI Codex extensively for candidate selection, conjecture formation, symbolic and finite-field derivations, adversarial proof checking, literature-search assistance, verification-code generation, and manuscript drafting and revision. The retained claims are exposed through human-readable arguments and reproducible artifacts, but AI output is not treated as an authority and independent expert review remains pending. Codex is a tool, not an author. The named author, Carptopus, accepts responsibility for the manuscript’s accuracy, originality, citations, and integrity.

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